📘 IoT PRACTICAL IMPLEMENTATIONS

🧪 1. Raspberry Pi Familiarization & Software Installation

🎯 Aim

To install OS and familiarize with Raspberry Pi.

⚙️ Procedure

1. Download Raspberry Pi Imager
2. Insert SD card
3. Select **Raspberry Pi OS**
4. Write OS to SD card
5. Insert SD card into Pi
6. Connect monitor, keyboard, power
7. Boot system

💻 Commands

```
sudo apt update
sudo apt upgrade -y
```

📌 Result

Raspberry Pi OS successfully installed and configured.

🧪 2. Word & Character Count Program (Python)

🎯 Aim

To count words and characters in a string using Python.

💻 Code

```
text = input("Enter a string: ")

words = len(text.split())
chars = len(text)

print("Word count:", words)
print("Character count:", chars)
```

📌 Result

Program successfully counts words and characters.

🧪 3. Raspberry Pi 40-PIN GPIO Study

🎯 Aim

To study 40-pin GPIO interface of Raspberry Pi.

📌 Key Pins:

- Power: 5V, 3.3V
- Ground: GND
- GPIO pins: Input/Output
- Communication: SPI, I2C, UART

⚙️ Working

GPIO pins are used to interface sensors, LEDs, motors, and other components.

🧪 4. Bluetooth Controlled LED using Raspberry Pi

🎯 Aim

To control LED using Bluetooth input.

⚙️ Components

- Bluetooth module
- LED + resistor

💻 Code

```
import serial
import RPi.GPIO as GPIO

GPIO.setmode(GPIO.BCM)
GPIO.setup(17, GPIO.OUT)

bluetooth = serial.Serial("/dev/rfcomm0", 9600)

while True:
    data = bluetooth.read().decode('utf-8')

    if data == '1':
        GPIO.output(17, True)
    elif data == '0':
        GPIO.output(17, False)
```

📌 Result

LED controlled via smartphone Bluetooth.

🧪 5. Connect Arduino with Wi-Fi Network

🎯 Aim

To connect Arduino Uno to Wi-Fi.

⚙️ Method

Use ESP8266 Wi-Fi module.

💻 Code (Basic)

```
#include <ESP8266WiFi.h>

const char* ssid = "WiFi_Name";
const char* password = "Password";

void setup() {
  Serial.begin(9600);
  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
  }

  Serial.println("Connected to WiFi");
}

void loop() {}
```

🧪 6. Raspberry Pi + ThingSpeak Temperature Data

🎯 Aim

To send temperature/humidity to cloud.

💻 Code

```
import requests
import Adafruit_DHT

sensor = Adafruit_DHT.DHT11
pin = 4

humidity, temperature = Adafruit_DHT.read_retry(sensor, pin)

url = "https://api.thingspeak.com/update?api_key=YOUR_API_KEY"
url += "&field1=" + str(temperature)

requests.get(url)
```

📌 Result

Data uploaded to cloud platform.

🧪 7. MQTT Publish Temperature Data

🎯 Aim

To publish sensor data using MQTT.

💻 Code

```
import paho.mqtt.client as mqtt
import time

client = mqtt.Client()
client.connect("broker.hivemq.com", 1883)
```

```
while True:
    temp = 25
    client.publish("iot/temperature", temp)
    time.sleep(5)
```

📌 Result

Temperature data published successfully.

🧪 8. Raspberry Pi System Integration

🎯 Aim

To interface multiple components.

⚙️ Example Components:

- LED
- Sensor
- Motor

📌 Working

GPIO pins connect hardware → Python controls logic → output devices respond.

🧪 9. DC Motor Speed Control using IoT

🎯 Aim

To control motor speed using PWM.

💻 Code (Raspberry Pi)

```
import RPi.GPIO as GPIO
import time

motor = 18
GPIO.setmode(GPIO.BCM)
GPIO.setup(motor, GPIO.OUT)

pwm = GPIO.PWM(motor, 100)
pwm.start(0)

pwm.ChangeDutyCycle(50)
time.sleep(5)

pwm.ChangeDutyCycle(100)
```

📌 Result

Motor speed controlled successfully.

🧪 10. IoT Data Logging using BeagleBone & ThingSpeak

🎯 Aim

To log sensor data in cloud using BeagleBone Black.

⚙️ Steps

1. Connect sensor
2. Read data using Python
3. Send data to ThingSpeak

💻 Code

```
import requests

data = 30
url = "https://api.thingspeak.com/update?api_key=YOUR_KEY&field1=" +
str(data)

requests.get(url)
```

📌 Result

Data logged successfully in cloud.